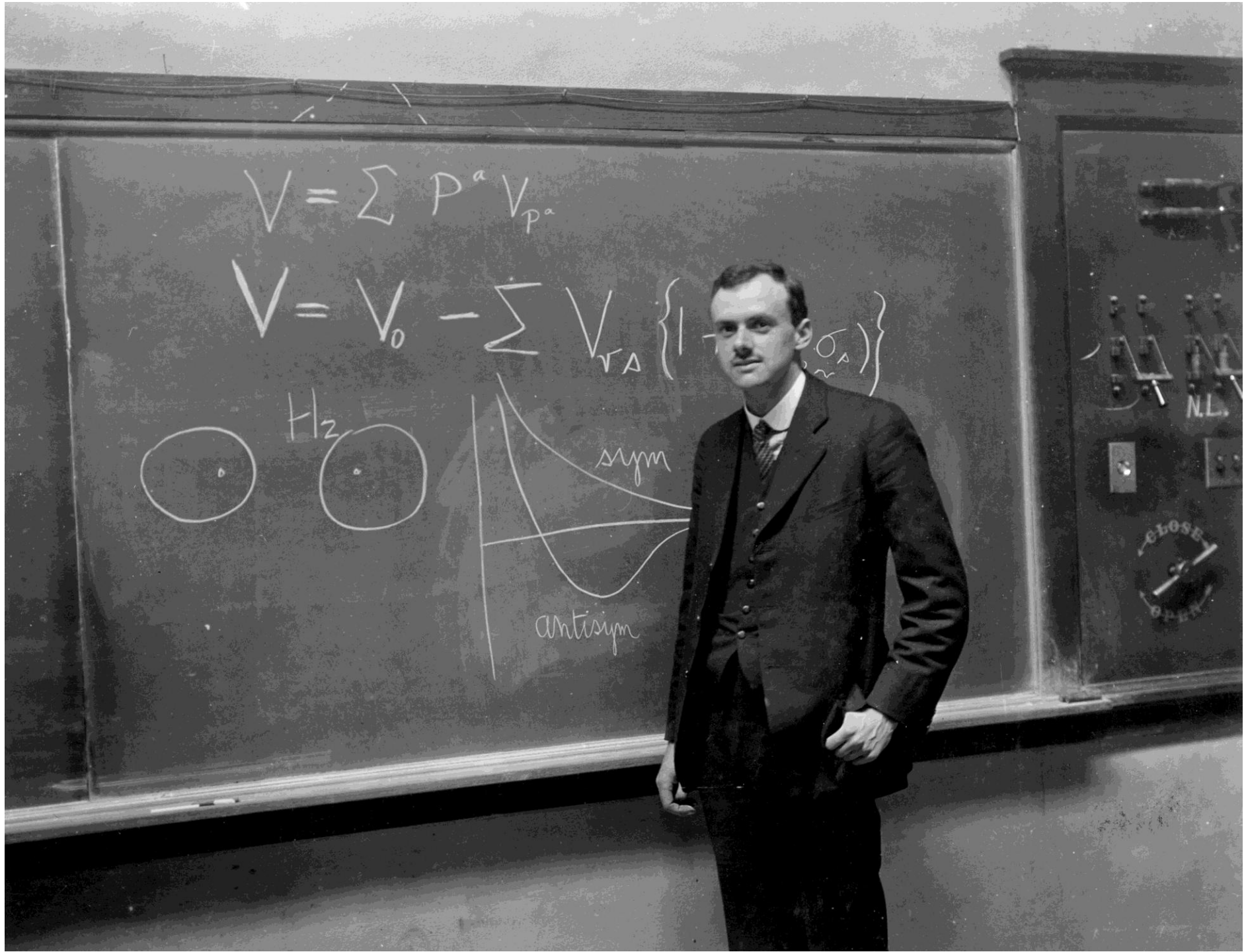

$V = \sum P^a V_{P^a}$
$V = V_0 - \sum V_{rs} \{ 1 - (\sigma_s \cdot \sigma_s) \}$
H_2
sym
antisym
N.L.
CLOSE

北京大學
PEKING UNIVERSITY

第三章 分子结构

2. 价键理论

（一） 氢分子薛定谔方程及价键法处理

第一次量子化学计算

- 1927年 海特勒和伦敦
- **近似**求解H_2薛定谔方程
- 获得能量曲线，发现基态成键!
- 定性正确，但与实验有差距

需要发展计算方法，**精确**求解薛定谔方程!

	键长 (Å)	键能 (eV)
HL	0.87	3.15
Exp	0.74	4.75

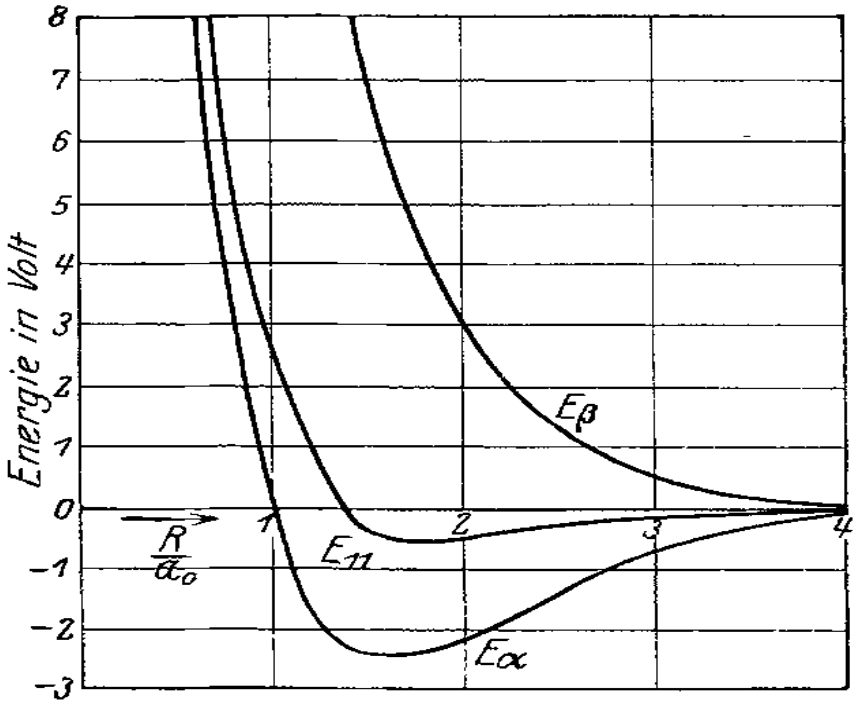
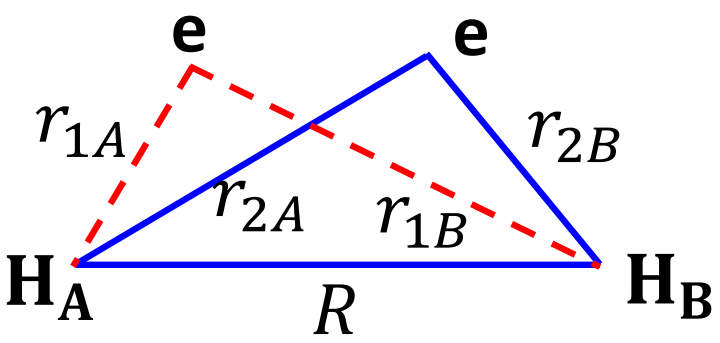


Fig. 1. Potential zweier neutraler H-Atome. (E_α = homöopolare Anziehung, E_β = elastische Reflexion.)

W. **Heitler**, F. **London** - Zeitschrift für Physik, 1927

H_2分子薛定谔方程



$$\left(-\frac{1}{2}\nabla_1^2-\frac{1}{2}\nabla_2^2\right)\Psi+\left[-\frac{1}{|\boldsymbol{r}_1-\boldsymbol{R}_A|}-\frac{1}{|\boldsymbol{r}_1-\boldsymbol{R}_B|}-\frac{1}{|\boldsymbol{r}_2-\boldsymbol{R}_A|}-\frac{1}{|\boldsymbol{r}_2-\boldsymbol{R}_B|}+\frac{1}{|\boldsymbol{r}_1-\boldsymbol{r}_2|}\right]\Psi+\frac{1}{R}\Psi=E\Psi$$

$$\Psi=\Psi(\boldsymbol{r}_1,s_1;\boldsymbol{r}_2,s_2)$$

$$=\Psi(x_1,y_1,z_1,s_1;x_2,y_2,z_2,s_2)$$

s_i是自旋维度， $s_i=\uparrow$ or $\downarrow$

$$\nabla_1^2=\left(\frac{\partial}{\partial x_1}\right)^2+\left(\frac{\partial}{\partial y_1}\right)^2+\left(\frac{\partial}{\partial z_1}\right)^2$$

$$\nabla_2^2=\left(\frac{\partial}{\partial x_2}\right)^2+\left(\frac{\partial}{\partial y_2}\right)^2+\left(\frac{\partial}{\partial z_2}\right)^2$$

- 电子的交换对称性

$$\Psi(\boldsymbol{r}_1,s_1;\boldsymbol{r}_2,s_2)=-\Psi(\boldsymbol{r}_2,s_2;\boldsymbol{r}_1,s_1)$$

又称Pauli不相容原理

- 薛定谔方程不含自旋维度，可进行变量分离

$$\Psi(\boldsymbol{r}_1, s_1; \boldsymbol{r}_2, s_2) = \Phi(\boldsymbol{r}_1, \boldsymbol{r}_2)\chi(s_1, s_2)$$

- 其中空间波函数满足薛定谔方程

$$(-\frac{1}{2}\nabla_1^2 - \frac{1}{2}\nabla_2^2)\Phi + \left[-\frac{1}{|\boldsymbol{r}_1 - \boldsymbol{R}_A|} - \frac{1}{|\boldsymbol{r}_1 - \boldsymbol{R}_B|} - \frac{1}{|\boldsymbol{r}_2 - \boldsymbol{R}_A|} - \frac{1}{|\boldsymbol{r}_2 - \boldsymbol{R}_B|} + \frac{1}{|\boldsymbol{r}_1 - \boldsymbol{r}_2|}\right]\Phi + \frac{1}{R}\Phi = E\Phi$$

- 由电子交换对称性，可得两种可能

(1) $\Phi(\boldsymbol{r}_1, \boldsymbol{r}_2) = \Phi(\boldsymbol{r}_2, \boldsymbol{r}_1), \qquad \chi(s_1, s_2) = -\chi(s_2, s_1)$

(2) $\Phi(\boldsymbol{r}_1, \boldsymbol{r}_2) = -\Phi(\boldsymbol{r}_2, \boldsymbol{r}_1), \qquad \chi(s_1, s_2) = \chi(s_2, s_1)$

(1) $\Phi(\boldsymbol{r}_1, \boldsymbol{r}_2) = \Phi(\boldsymbol{r}_2, \boldsymbol{r}_1), \qquad \chi(s_1, s_2) = -\chi(s_2, s_1)$

$$\chi^A(s_1, s_2) = \frac{1}{\sqrt{2}}\Big(\alpha(1)\beta(2) - \alpha(2)\beta(1)\Big) = \frac{1}{\sqrt{2}}\Big(1_\uparrow 2_\downarrow - 2_\uparrow 1_\downarrow\Big)$$

只有一种形式，对应单重态

思考题：单重态与三重态，能量孰高？

(2) $\Phi(\boldsymbol{r}_1, \boldsymbol{r}_2) = -\Phi(\boldsymbol{r}_2, \boldsymbol{r}_1), \qquad \chi(s_1, s_2) = \chi(s_2, s_1)$

$$\chi^S(s_1, s_2) = \begin{cases} \alpha(1)\alpha(2) \\ \frac{1}{\sqrt{2}}\Big(\alpha(1)\beta(2) + \alpha(2)\beta(1)\Big) \\ \beta(1)\beta(2) \end{cases}$$

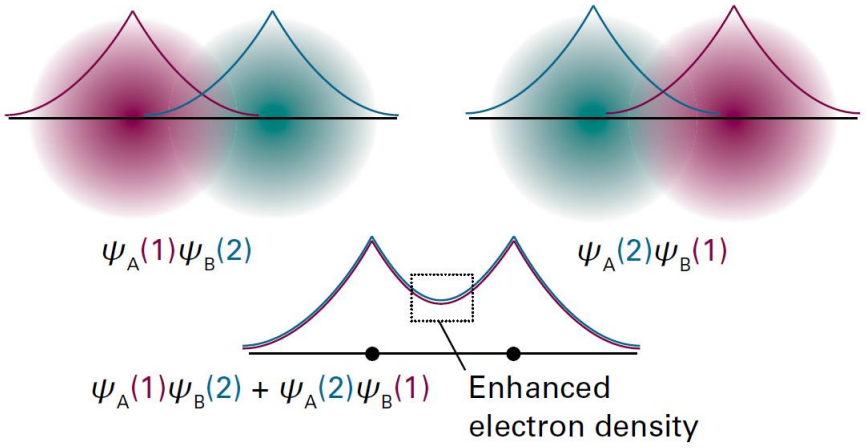
三种形式，对应三重态

近似求解空间波函数

$$\psi_A(\boldsymbol{r}) = \psi_{1s}(\boldsymbol{r} - \boldsymbol{R}_A) = \frac{1}{\sqrt{\pi}} e^{-|r-\boldsymbol{R}_A|}$$

- 基本思想：用H原子波函数构建氢分子波函数

$$\Phi(\boldsymbol{r}_1, \boldsymbol{r}_2) = c_1 \psi_A(\boldsymbol{r}_1)\psi_B(\boldsymbol{r}_2) + c_2 \psi_A(\boldsymbol{r}_2)\psi_B(\boldsymbol{r}_1)$$



- 简写为

$$\Phi(1,2) = c_1 \psi_A(1)\psi_B(2) + c_2 \psi_A(2)\psi_B(1)$$

- 问题：Φ是否满足薛定谔方程？

$$\hat{H}\Phi \overset{?}{=} E\Phi$$

- 不满足，是价键法近似，c_1, c_2为待求参数

对称性决定：
单重态$c_1 = c_2$，
三重态$c_1 = -c_2$

- c_1, c_2的大小通过波函数归一化得到

$$\int |\Phi(\boldsymbol{r}_1, \boldsymbol{r}_2)|^2 d\boldsymbol{r}_1 d\boldsymbol{r}_2 = 1$$

$$\Longrightarrow \int |c_1\psi_A(\boldsymbol{r}_1)\psi_B(\boldsymbol{r}_2) + c_2\psi_A(\boldsymbol{r}_2)\psi_B(\boldsymbol{r}_1)|^2 d\boldsymbol{r}_1 d\boldsymbol{r}_2 = 1$$

$$\Longrightarrow c_1^2(2 \pm 2S_{12}) = 1$$

$$S_{12} = \int \psi_A(\boldsymbol{r}_1)\psi_B(\boldsymbol{r}_2)\psi_A(\boldsymbol{r}_2)\psi_B(\boldsymbol{r}_1) d\boldsymbol{r}_1 d\boldsymbol{r}_2$$

单重态（基态）

$$c_1 = c_2 = \frac{1}{\sqrt{2 + 2S_{12}}}$$

$$\Phi_S = \frac{\psi_A(\boldsymbol{r}_1)\psi_B(\boldsymbol{r}_2) + \psi_A(\boldsymbol{r}_2)\psi_B(\boldsymbol{r}_1)}{\sqrt{2 + 2S_{12}}}$$

三重态（激发态）

$$c_1 = -c_2 = \frac{1}{\sqrt{2 - 2S_{12}}}$$

$$\Phi_A = \frac{\psi_A(\boldsymbol{r}_1)\psi_B(\boldsymbol{r}_2) - \psi_A(\boldsymbol{r}_2)\psi_B(\boldsymbol{r}_1)}{\sqrt{2 - 2S_{12}}}$$

- 改写能量表达式，得到

$$E = \frac{1}{R} + \int \Phi^*(\boldsymbol{r}_1, \boldsymbol{r}_2)\hat{H}\Phi(\boldsymbol{r}_1, \boldsymbol{r}_2)d\boldsymbol{r}_1 d\boldsymbol{r}_2$$

$$\hat{H} = -\frac{1}{2}\nabla_1^2 - \frac{1}{2}\nabla_2^2 + \left[-\frac{1}{|\boldsymbol{r}_1 - \boldsymbol{R}_A|} - \frac{1}{|\boldsymbol{r}_1 - \boldsymbol{R}_B|} - \frac{1}{|\boldsymbol{r}_2 - \boldsymbol{R}_A|} - \frac{1}{|\boldsymbol{r}_2 - \boldsymbol{R}_B|}\right] + \frac{1}{|\boldsymbol{r}_1 - \boldsymbol{r}_2|}$$

- 代入波函数表达式，改写得

$$E = \frac{1}{R} + H_{11}c_1^2 + H_{22}c_2^2 + 2H_{12}c_1c_2$$

$$H_{11} = \int \psi_A(1)\psi_B(2)\ \hat{H}\ \Big(\psi_A(1)\psi_B(2)\Big)d\boldsymbol{r}_1 d\boldsymbol{r}_2$$

$$H_{22} = \int \psi_A(2)\psi_B(1)\ \hat{H}\ \Big(\psi_A(2)\psi_B(1)\Big)d\boldsymbol{r}_1 d\boldsymbol{r}_2$$

$$H_{12} = \int \psi_A(1)\psi_B(2)\ \hat{H}\ \Big(\psi_A(2)\psi_B(1)\Big)d\boldsymbol{r}_1 d\boldsymbol{r}_2$$

由对称性知

$$H_{11} = H_{22}$$

代入c_1, c_2得

$$E = \frac{1}{R} + \frac{H_{11} \pm H_{12}}{1 \pm S_{12}}$$

（二） 双电子积分

能量表达式中各项分析

$$S_{12} = \int \psi_A(\boldsymbol{r}_1)\psi_B(\boldsymbol{r}_2)\psi_A(\boldsymbol{r}_2)\psi_B(\boldsymbol{r}_1)d\boldsymbol{r}_1 d\boldsymbol{r}_2$$

$$= \left(\int \psi_A(\boldsymbol{r}_1)\psi_B(\boldsymbol{r}_1)d\boldsymbol{r}_1\right)\left(\int \psi_A(\boldsymbol{r}_2)\psi_B(\boldsymbol{r}_2)d\boldsymbol{r}_2\right)$$

$$= S_{AB}^2$$

r_1 and r_2 are dummy variables

$$S_{AB} = \int \psi_A(\boldsymbol{r})\psi_B(\boldsymbol{r})d\boldsymbol{r}$$

重叠积分

S_{12}是双电子波函数层面的重叠积分；
S_{ab}是单电子波函数（轨道）层面的重叠积分

$$\hat{H} = -\frac{1}{2}\nabla_1^2 - \frac{1}{2}\nabla_2^2 + \left[-\frac{1}{|\boldsymbol{r}_1 - \boldsymbol{R}_A|} - \frac{1}{|\boldsymbol{r}_1 - \boldsymbol{R}_B|} - \frac{1}{|\boldsymbol{r}_2 - \boldsymbol{R}_A|} - \frac{1}{|\boldsymbol{r}_2 - \boldsymbol{R}_B|} \right] + \frac{1}{|\boldsymbol{r}_1 - \boldsymbol{r}_2|}$$

- 对能量部分进行分组

$$\hat{H}_A(\boldsymbol{r}_1) = -\frac{1}{2}\nabla_1^2 - \frac{1}{|\boldsymbol{r}_1 - \boldsymbol{R}_A|} \qquad \hat{H}_B(\boldsymbol{r}_2) = -\frac{1}{2}\nabla_2^2 - \frac{1}{|\boldsymbol{r}_2 - \boldsymbol{R}_B|}$$

$$\hat{H}_{AB}(\boldsymbol{r}_1, \boldsymbol{r}_2) = -\frac{1}{|\boldsymbol{r}_1 - \boldsymbol{R}_B|} - \frac{1}{|\boldsymbol{r}_2 - \boldsymbol{R}_A|} + \frac{1}{|\boldsymbol{r}_1 - \boldsymbol{r}_2|}$$

$$\hat{H} = \hat{H}_A + \hat{H}_B + \hat{H}_{AB}$$

$$H_{11} = \int \psi_A(1)\psi_B(2)\ \hat{H}\ \Big(\psi_A(1)\psi_B(2)\Big) d\boldsymbol{r}_1 d\boldsymbol{r}_2$$

$$H_{12} = \int \psi_A(1)\psi_B(2)\ \hat{H}\ \Big(\psi_A(2)\psi_B(1)\Big) d\boldsymbol{r}_1 d\boldsymbol{r}_2$$

$$H_{11} = \int \psi_A(1)\psi_B(2)\ (\hat{H}_A + \hat{H}_B + \hat{H}_{AB})\ \Big(\psi_A(1)\psi_B(2)\Big) d\boldsymbol{r}_1 d\boldsymbol{r}_2$$

$$\int \psi_A(1)\psi_B(2)\ \hat{H}_A\ \Big(\psi_A(1)\psi_B(2)\Big) d\boldsymbol{r}_1 d\boldsymbol{r}_2$$

$$= \int \psi_A(1)\psi_B(2)\ \Big(\hat{H}_A\ \psi_A(1)\Big)\psi_B(2) d\boldsymbol{r}_1 d\boldsymbol{r}_2$$

$$= \int \psi_A(1)\ \Big(\hat{H}_A\ \psi_A(1)\Big) d\boldsymbol{r}_1 \int \psi_B(2)\psi_B(2) d\boldsymbol{r}_2$$

$$= -\frac{1}{2}$$

$$\int \psi_A(1)\psi_B(2)\ \hat{H}_B\ \Big(\psi_A(1)\psi_B(2)\Big) d\boldsymbol{r}_1 d\boldsymbol{r}_2$$

$$= \int \psi_A(1)\psi_B(2)\ \Big(\hat{H}_B\ \psi_B(2)\Big)\psi_A(1) d\boldsymbol{r}_1 d\boldsymbol{r}_2$$

$$= \int \psi_A(1)\psi_A(1) d\boldsymbol{r}_1 \int \psi_B(2)\ \Big(\hat{H}_B\ \psi_B(2)\Big) d\boldsymbol{r}_2$$

$$= -\frac{1}{2}$$

$$\int \psi_A(1)\psi_B(2)\ \hat{H}_{AB}\ \Big(\psi_A(1)\psi_B(2)\Big) d\boldsymbol{r}_1 d\boldsymbol{r}_2$$

$$= \int |\psi_A(1)|^2 \Big(-\frac{1}{|\boldsymbol{r}_1 - \boldsymbol{R}_2|} - \frac{1}{|\boldsymbol{r}_2 - \boldsymbol{R}_1|} + \frac{1}{|\boldsymbol{r}_1 - \boldsymbol{r}_2|} \Big) |\psi_B(2)|^2 d\boldsymbol{r}_1 d\boldsymbol{r}_2$$

$$\equiv Q$$

库伦积分

$$H_{12} = \int \psi_A(1)\psi_B(2)\ (\hat{H}_A + \hat{H}_B + \hat{H}_{AB})\ \Big(\psi_A(2)\psi_B(1)\Big) d\boldsymbol{r}_1 d\boldsymbol{r}_2$$

$$\int \psi_A(1)\psi_B(2)\ \hat{H}_A\ \Big(\psi_A(2)\psi_B(1)\Big) d\boldsymbol{r}_1 d\boldsymbol{r}_2$$

$$= \int \psi_A(1)\psi_B(2)\ \Big(\hat{H}_A\ \psi_B(1)\Big)\psi_A(2) d\boldsymbol{r}_1 d\boldsymbol{r}_2$$

$$= \int \psi_A(1)\ \Big(\hat{H}_A\ \psi_B(1)\Big) d\boldsymbol{r}_1 \int \psi_A(2)\psi_B(2) d\boldsymbol{r}_2$$

$$= -\frac{1}{2} S_{AB}^2 \qquad \int \psi_B(1)\ \Big(\hat{H}_A\ \psi_A(1)\Big) d\boldsymbol{r}_1$$

$$\int \psi_A(1)\psi_B(2)\ \hat{H}_B\ \Big(\psi_A(2)\psi_B(1)\Big) d\boldsymbol{r}_1 d\boldsymbol{r}_2$$

$$= \int \psi_A(1)\psi_B(2)\ \Big(\hat{H}_B\ \psi_A(2)\Big)\psi_B(1) d\boldsymbol{r}_1 d\boldsymbol{r}_2$$

$$= \int \psi_A(1)\psi_B(1) d\boldsymbol{r}_1 \int \psi_B(2)\ \Big(\hat{H}_B\ \psi_A(2)\Big) d\boldsymbol{r}_2$$

$$= -\frac{1}{2} S_{AB}^2$$

$$\int \psi_A(1)\psi_B(2)\ \hat{H}_{AB}\ \Big(\psi_A(2)\psi_B(1)\Big) d\boldsymbol{r}_1 d\boldsymbol{r}_2$$

$$= \int \psi_A(1)\psi_B(1)\Big(-\frac{1}{|\boldsymbol{r}_1 - \boldsymbol{R}_2|} - \frac{1}{|\boldsymbol{r}_2 - \boldsymbol{R}_1|} + \frac{1}{|\boldsymbol{r}_1 - \boldsymbol{r}_2|}\Big)\psi_A(2)\psi_B(2) d\boldsymbol{r}_1 d\boldsymbol{r}_2$$

$$\equiv A$$

交换积分

化简库伦积分

$$Q = \int |\psi_A(1)|^2 \left(-\frac{1}{|\boldsymbol{r}_1 - \boldsymbol{R}_B|} - \frac{1}{|\boldsymbol{r}_2 - \boldsymbol{R}_A|} + \frac{1}{|\boldsymbol{r}_1 - \boldsymbol{r}_2|} \right) |\psi_B(2)|^2 d\boldsymbol{r}_1 d\boldsymbol{r}_2$$

$$\int |\psi_A(1)|^2 \left(-\frac{1}{|\boldsymbol{r}_1 - \boldsymbol{R}_B|} \right) |\psi_B(2)|^2 d\boldsymbol{r}_1 d\boldsymbol{r}_2 = -\int \frac{|\psi_A(\boldsymbol{r}_1)|^2}{|\boldsymbol{r}_1 - \boldsymbol{R}_B|} d\boldsymbol{r}_1 = -\int \frac{|\psi_A(\boldsymbol{r})|^2}{|\boldsymbol{r} - \boldsymbol{R}_B|} d\boldsymbol{r} \equiv -P$$

$$\int |\psi_A(1)|^2 \left(-\frac{1}{|\boldsymbol{r}_2 - \boldsymbol{R}_A|} \right) |\psi_B(2)|^2 d\boldsymbol{r}_1 d\boldsymbol{r}_2 = -\int \frac{|\psi_B(\boldsymbol{r}_2)|^2}{|\boldsymbol{r}_2 - \boldsymbol{R}_A|} d\boldsymbol{r}_2 = -\int \frac{|\psi_B(\boldsymbol{r})|^2}{|\boldsymbol{r} - \boldsymbol{R}_A|} d\boldsymbol{r} \equiv -P$$

双电子库伦积分

$$J = \int |\psi_A(1)|^2 \frac{1}{|\boldsymbol{r}_1 - \boldsymbol{r}_2|} |\psi_B(2)|^2 d\boldsymbol{r}_1 d\boldsymbol{r}_2$$

化简交换积分

$$G = \int \frac{\psi_A(\boldsymbol{r})\psi_B(\boldsymbol{r})}{|\boldsymbol{r}-\boldsymbol{R}_A|}d\boldsymbol{r} = \int \frac{\psi_A(\boldsymbol{r})\psi_B(\boldsymbol{r})}{|\boldsymbol{r}-\boldsymbol{R}_B|}d\boldsymbol{r}$$

$$A = \int \psi_A(1)\psi_B(1)\left(-\frac{1}{|\boldsymbol{r}_1-\boldsymbol{R}_B|} - \frac{1}{|\boldsymbol{r}_2-\boldsymbol{R}_A|} + \frac{1}{|\boldsymbol{r}_1-\boldsymbol{r}_2|}\right)\psi_A(2)\psi_B(2)d\boldsymbol{r}_1 d\boldsymbol{r}_2$$

$$\int \psi_A(1)\psi_B(1)\left(-\frac{1}{|\boldsymbol{r}_1-\boldsymbol{R}_B|}\right)\psi_A(2)\psi_B(2)d\boldsymbol{r}_1 d\boldsymbol{r}_2 = -S_{AB}\int \frac{\psi_A(1)\psi_B(1)}{|\boldsymbol{r}_1-\boldsymbol{R}_B|}d\boldsymbol{r}_1 \equiv -S_{AB}G$$

$$\int \psi_A(1)\psi_B(1)\left(-\frac{1}{|\boldsymbol{r}_2-\boldsymbol{R}_A|}\right)\psi_A(2)\psi_B(2)d\boldsymbol{r}_1 d\boldsymbol{r}_2 = -S_{AB}\int \frac{\psi_A(2)\psi_B(2)}{|\boldsymbol{r}_2-\boldsymbol{R}_A|}d\boldsymbol{r}_1 \equiv -S_{AB}G$$

双电子交换积分

$$K = \int \psi_A(1)\psi_B(1)\frac{1}{|\boldsymbol{r}_1-\boldsymbol{r}_2|}\psi_A(2)\psi_B(2)d\boldsymbol{r}_1 d\boldsymbol{r}_2$$

- 计算双电子积分常常是量化计算的决速步

（三） 结果分析及改进

- 综合以上结果

$$H_{11} = -1 + Q, \qquad H_{12} = -S_{AB}^2 + A$$

$$E_1 = \frac{1}{R} + \frac{H_{11} + H_{12}}{1 + S_{12}} = \frac{1}{R} - 1 + \frac{Q + A}{1 + S_{AB}^2}$$

$$E_2 = \frac{1}{R} + \frac{H_{11} - H_{12}}{1 - S_{12}} = \frac{1}{R} - 1 + \frac{Q - A}{1 - S_{AB}^2}$$

- 主要的能量稳定（成键能）贡献来自交换积分A
  - 假如没有A，则无法成键

- 交换积分A是纯粹的非经典项，库伦积分是经典项

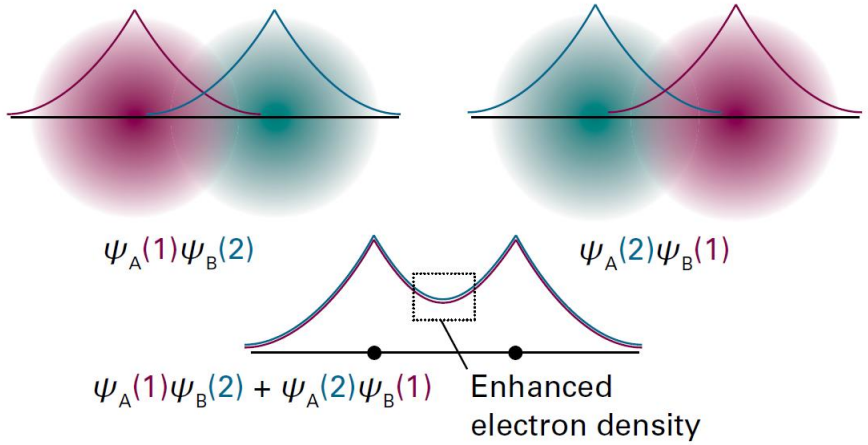
改进方案

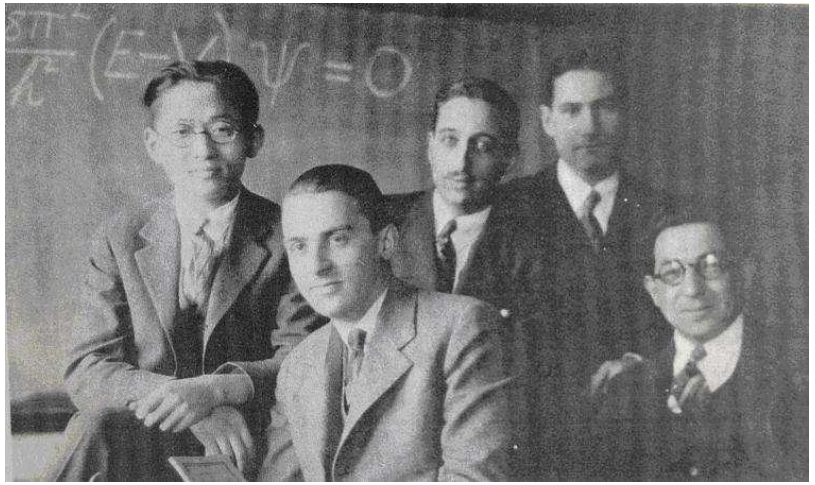
- 修改原子轨道，引入变分参数 Z

$$\psi_A(\boldsymbol{r}) = \frac{Z^{3/2}}{\sqrt{\pi}} e^{-Z|\boldsymbol{r}-\boldsymbol{R}_A|}$$

$$\Phi(\boldsymbol{r}_1, \boldsymbol{r}_2) = c_1 \psi_A(\boldsymbol{r}_1)\psi_B(\boldsymbol{r}_2) + c_2 \psi_A(\boldsymbol{r}_2)\psi_B(\boldsymbol{r}_1)$$

- 重复Heitler-London的计算步骤，得 $E = E(Z)$
- 对$E(Z)$求极小值，得到基态能量的近似





左一：王守竞

其他改进方案

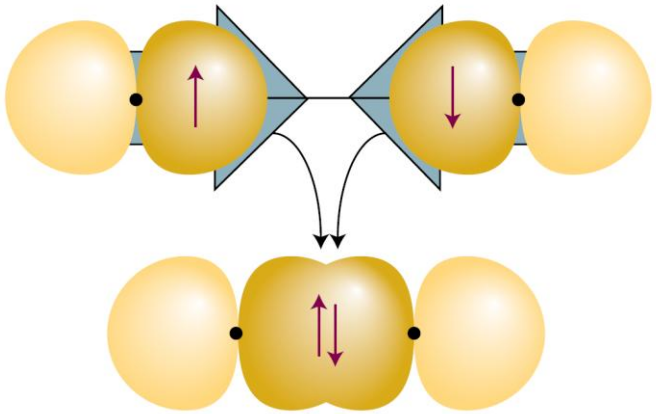
Author	Year	method	R_e (Å)	D_e (eV)
Coulson	1937	LCAO MO	0.732	3.49
Kolos and Roothaan	1960	Hartree-Fock	0.732	3.64
Weinbaum	1933	CI (1 para)	0.757	4.03
Hagstrom and Shull	1963	CI (33 para)	-	4.71
James and Coolidge	1933	Variational (13 para)	-	4.72
		Exp	0.7414	4.75
Author	Year	method	R_e (Å)	D_0/hc (cm^{-1})
Kolos, Wolniewicz et al.	1986	Variational (279 para)	-	36118.1
Piszczatowski et al.	2009	Variational (7000 para)		36118.0695(10)
		Exp	0.7414	36118.0696(4)

总结

- Heitler-London价键法求解H_2 能量曲线
- 求解方法：

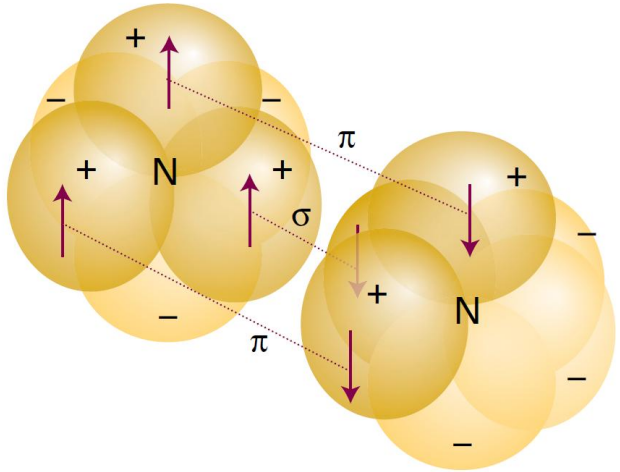
  原子轨道构建双电子波函数，进而组合成符合对称性的双电子波函数，该法又称价键法
- 基态能量曲线有极小值点，对应平衡键长
- **自旋配对**对成键起到关键作用
- 其他方法：如分子轨道法
- 思考：价键法与分子轨道法的异同（作业题）

多电子原子的价键理论



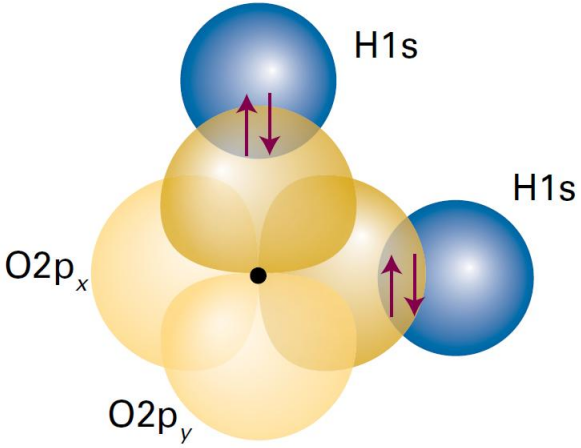
- 共振（resonance）
- 跃迁（promotion）
- 杂化（hybridization）
- 杂化轨道（hybrid orbital）

$$\Psi_{SB} = C\Big(\psi_A(\boldsymbol{r}_1)\psi_B(\boldsymbol{r}_2) + \psi_A(\boldsymbol{r}_2)\psi_B(\boldsymbol{r}_1)\Big)\frac{1}{\sqrt{2}}\Big(\alpha(1)\beta(2) - \alpha(2)\beta(1)\Big)$$



以上内容阅读教材自学

思考：在价键理论下多电子全波函数怎么写？



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